Well Testing

A HIPAA-Aware Benchmark and Evaluation Harness for Clinical LLMs to Quantify Hallucination, Bias, and PHI Leakage

Valentina Palama
MSc in Computer Information Systems
(Prairie View A&M University), USA
Email: Valentina.palama@yahoo.com

Abstract
The growing use of large language models (LLMs) in clinical practice has brought up serious doubts about the reliability, fairness, and compliance with regulations. Model hallucinations may undermine clinical decision-making in a health care setting, algorithmic bias may create health inequities, and unintended disclosure of protected health information (PHI), may contravene privacy rules. Although more focus is given to clinical LLM assessment, current benchmarks pay much attention to overall performance and do not address these safety and compliance risks in a holistic manner in a HIPAA-conscious system. This paper suggests a uniform benchmark and assessment harness that is specifically rigorously developed in clinical LLMs to quantify hallucination, bias, and leakage of PHI systematically. The framework includes clinically-based exercises, de-identified and artificial data sets, and automated identification tools consistent with the HIPAA-related privacy categories. Through the combination of several dimensions of evaluation into a single and reproducible harness, the benchmark can allow comparative clinical LLM evaluation of safety, fairness and privacy measures. The findings demonstrate that there is a significant variability in model behavior, which indicates trade-offs between clinical capability and risk exposure. The work adds a useful evaluation system to help implement LLMs responsibly, regulate, and monitor their use in healthcare settings.

Keywords:
Clinical large language models; HIPAA compliance; hallucination detection; algorithmic bias; protected health information leakage; healthcare AI evaluation

1. Introduction

Large language models (LLMs) are finding their way into clinical and healthcare processes, such as analysis of electronic health records (EHRs), medical question-answering systems, clinical documentation, decision support systems, and patient-facing, among others. This has enabled them to be core building blocks in the new data-centric designs of healthcare because of their capabilities to process unstructured clinical text and produce human-like responses (Zhang et al., 2024; Du et al., 2024). The systematic reviews in the field of clinical, mental health areas indicate the potential of LLMs transforming healthcare but also have significant risks that are inherent to their use in the context of sensitive healthcare (Guo et al., 2024; Kamath et al., 2024).

Clinical LLMs, in spite of their promising performance, have continued to have issues of safety and trustworthiness. The most serious of them is the occurrence of hallucination, or the creation of fluent yet inaccurate or clinically unsafe content, which has been extensively recorded in EHR-related activities and medical question answering systems (Du et al., 2024; Pham and Vo, 2024; Ahmad et al., 2023). Such errors can spread misinformation, interfere with the trust of clinicians and directly endanger patient safety in clinical settings. To reduce the risk of hallucination, there are concerns of algorithmic bias, in which the results of LLM can reproduce or amplify demographic, socioeconomic, or clinical disparities, which in turn worsen existing health inequities (Rani et al., 2024).

The problem of patient privacy protection is also important. Clinical LLMs that are trained or tested on structured health data can knowingly or unknowingly memorize or reproduce sensitive health information (PHI), which will be highly uncompliant with laws like the Health Insurance Portability and Accountability Act (HIPAA). The earlier research has quantitatively observed the presence of leakage vulnerability of clinical language models to templates prompts or adversarial query methods (Toval Borrell & Flores, 2024). Similar studies in personal health libraries and HIPAA-sensitive systems highlight the relevance of formal privacy-aware systems and assessment systems in digital health technologies (Jamill et al., 2021; Jamil et al., 2021a; Jamil, 2021; Ammar et al., 2020).

Although a number of assessment frameworks and clinical LLM benchmarks have been introduced recently, the majority target individual aspects, including accuracy of the task or overall robustness. Complete systems such as MEDIC offer useful multi-task clinical assessment and explicitly lack regulatory privacy limits or standard PHI leakage measurements (Kanithi et al., 2024). On the same note, more extensive hallucination-oriented assessment interventions suggest safety taxonomies and general verification systems but are not adequately adjusted to regulatory and ethical demands of medical settings (Liu et al., 2024). As a result, there is a lack of unified, HIPAA-aware benchmarking infrastructures capable of simultaneously assessing hallucination, bias, and PHI leakage in clinical LLMs.

This gap presents a critical barrier to responsible clinical adoption, regulatory oversight, and comparative model assessment. Addressing these challenges requires an evaluation harness that is clinically grounded, privacy-aware, and methodologically reproducible. By embedding HIPAA-relevant considerations directly into benchmark design and metric construction, such a framework can provide actionable insights into the trade-offs between model capability, fairness, and privacy preservation. This study is motivated by the need to move beyond fragmented evaluations toward an integrated, compliance-oriented approach that supports safe, trustworthy, and accountable deployment of LLMs in healthcare systems.

2. Background and Related Work

The rapid advancement of large language models (LLMs) has accelerated their adoption across a wide range of clinical and healthcare applications, including electronic health record (EHR) summarization, clinical decision support, medical question answering, and patient-facing tools. Recent systematic reviews highlight that foundation models adapted to healthcare contexts demonstrate promising performance gains, yet remain constrained by safety, reliability, and compliance challenges that are particularly acute in regulated clinical environments (Du et al., 2024; Zhang et al., 2024). As a result, evaluation of clinical LLMs has become a central research focus, extending beyond accuracy toward trustworthiness dimensions such as hallucination, bias, and privacy preservation.

Hallucination—defined as the generation of factually incorrect or clinically unsupported content—has been consistently identified as a major barrier to safe deployment of LLMs in healthcare. Prior studies have examined hallucination mechanisms in medical question answering and EHR-related tasks, noting that even domain-adapted models can fabricate diagnoses, treatments, or patient attributes when confronted with ambiguous or underspecified prompts (Ahmad et al., 2023; Pham & Vo, 2024). Emerging evaluation frameworks seek to quantify hallucination through task-specific benchmarks and uncertainty-aware metrics, yet these efforts often lack explicit alignment with clinical risk severity or regulatory implications (Liu et al., 2024). Moreover, many hallucination benchmarks remain general-purpose, limiting their applicability to real-world clinical documentation and reasoning workflows.

In parallel, algorithmic bias in healthcare LLMs has gained increasing attention due to its potential to amplify existing health disparities. Empirical studies demonstrate that LLM outputs may vary systematically across demographic attributes such as race, gender, and socioeconomic status, affecting diagnostic suggestions, triage recommendations, and mental health assessments (Rani et al., 2024; Guo et al., 2024). While mitigation strategies including data balancing, prompt engineering, and post-hoc calibration have been proposed, standardized evaluation protocols for measuring bias in clinical contexts remain underdeveloped. Existing work often evaluates bias in isolation, without considering its interaction with hallucination or privacy risks (Kamath et al., 2024).

Privacy and confidentiality concerns represent an additional, and uniquely regulated, dimension of clinical LLM evaluation. Healthcare data are subject to strict legal protections, and inadvertent leakage of protected health information (PHI) poses significant ethical and legal risks. Prior research on PHI leakage has explored memorization and regurgitation risks in clinical language models trained on datasets such as MIMIC-III, demonstrating that template-based probing can elicit sensitive patient information under certain conditions (Toval Borrell & Flores, 2024). However, most leakage studies focus on isolated attack scenarios rather than systematic, benchmark-driven assessment. Earlier work on HIPAA-aware systems and personal health libraries underscores the importance of embedding privacy constraints directly into system architectures and evaluation processes (Jamill et al., 2021; Jamil et al., 2021; Jamil, 2021; Ammar et al., 2020).

Recent efforts have begun to move toward more comprehensive evaluation paradigms for clinical LLMs. Frameworks such as MEDIC propose multidimensional assessment across clinical performance, robustness, and safety, reflecting growing recognition that single-metric evaluations are insufficient for healthcare AI (Kanithi et al., 2024). Nevertheless, these frameworks typically stop short of explicitly operationalizing HIPAA-aligned privacy categories or integrating PHI leakage detection alongside hallucination and bias metrics. As highlighted in broader surveys of trustworthy and data-centric healthcare AI, there remains a critical gap in standardized, reproducible benchmarks that unify clinical validity, fairness, and regulatory compliance within a single evaluation harness (Zhang et al., 2024; Siuly et al., 2021).

Overall, existing literature establishes a strong foundation for understanding individual risks associated with clinical LLMs but reveals fragmentation across evaluation dimensions. This gap motivates the development of a HIPAA-aware benchmark and evaluation harness that jointly quantifies hallucination, bias, and PHI leakage, enabling more rigorous and policy-relevant assessment of clinical LLM safety and trustworthiness.

3. Problem Statement and Research Objectives

The rapid integration of large language models (LLMs) into clinical and healthcare applications has outpaced the development of rigorous, domain-specific evaluation frameworks capable of addressing safety, fairness, and regulatory compliance simultaneously. While recent studies demonstrate the potential of LLMs in electronic health record (EHR) processing, medical question answering, and clinical decision support, they also consistently report risks related to hallucinated content, biased outputs, and unreliable reasoning that may adversely affect patient care (Du et al., 2024; Pham & Vo, 2024; Ahmad et al., 2023). These challenges are further compounded by the sensitive nature of clinical data, where inadvertent disclosure of protected health information (PHI) constitutes not only a technical failure but also a regulatory and ethical violation.

Existing evaluation efforts primarily assess isolated dimensions of LLM performance, such as factual correctness or general trustworthiness, without explicitly incorporating HIPAA-aware privacy constraints or systematically quantifying PHI leakage (Kanithi et al., 2024; Liu et al., 2024). Although recent work has begun to explore hallucination mitigation and bias detection in healthcare LLMs, these approaches remain fragmented and lack standardized benchmarks tailored to realistic clinical tasks and compliance requirements (Rani et al., 2024; Kamath et al., 2024). Moreover, data-centric surveys highlight that evaluation pipelines often overlook how training data composition and prompting strategies influence hallucination, bias, and privacy risks in downstream clinical use (Zhang et al., 2024; Guo et al., 2024).

From a privacy and compliance perspective, prior research on HIPAA-aware systems and personal health libraries underscores the necessity of explicit architectural and evaluative mechanisms to prevent unauthorized PHI exposure (Jamill et al., 2021; Jamil et al., 2021; Jamil, 2021; Ammar et al., 2020). However, these principles have not been sufficiently operationalized within modern LLM benchmarking practices. Studies examining clinical language models using real-world datasets, such as MIMIC-III, further demonstrate that even de-identified data can lead to measurable information leakage under certain prompting conditions (Toval Borrell & Flores, 2024), reinforcing the need for systematic leakage detection as a first-class evaluation objective.

Against this backdrop, the core problem addressed in this study is the absence of an integrated, HIPAA-aware benchmarking and evaluation harness that can concurrently measure hallucination, bias, and PHI leakage in clinical LLMs using standardized, reproducible methodologies. Without such a framework, stakeholders lack the empirical evidence needed to compare models, understand risk trade-offs, and make informed decisions about clinical deployment.

Accordingly, the primary research objectives of this work are to:

1. define clinically grounded and HIPAA-relevant evaluation criteria for hallucination, bias, and PHI leakage;
2. develop a unified benchmark incorporating de-identified and synthetic clinical tasks that reflect real-world usage scenarios;
3. design an automated evaluation harness capable of detecting and quantifying these risks across multiple clinical LLMs; and
4. enable comparative, reproducible assessment to support responsible model selection, regulatory oversight, and continuous safety monitoring in healthcare AI systems.

4. Design of the HIPAA-Aware Benchmark

The HIPAA-aware benchmark is designed to provide a rigorous, clinically grounded, and regulation-sensitive framework for evaluating large language models deployed in healthcare contexts. Its design explicitly integrates three high-risk dimensions: hallucination, bias, and protected health information (PHI) leakage within a unified evaluation structure. This approach responds to gaps identified in prior clinical LLM evaluations, which often assess model accuracy or reasoning in isolation without sufficient attention to privacy compliance and trustworthiness (Du et al., 2024; Kanithi et al., 2024).

4.1 Design Principles and Scope

The benchmark is guided by four core principles. First, clinical realism ensures that tasks, prompts, and outputs reflect authentic healthcare workflows such as clinical question answering, discharge summary generation, and decision support, as emphasized in data-centric and task-oriented evaluations of healthcare LLMs (Zhang et al., 2024; Guo et al., 2024). Second, HIPAA alignment operationalizes privacy requirements by explicitly modeling PHI categories and risk levels, drawing on prior HIPAA-aware system architectures and personal health library frameworks (Jamill et al., 2021; Jamil et al., 2021). Third, multidimensional safety assessment integrates hallucination, bias, and PHI leakage into a single benchmark, addressing limitations of fragmented evaluation practices noted in existing literature (Pham & Vo, 2024; Rani et al., 2024). Finally, reproducibility and extensibility are prioritized through standardized datasets, modular task definitions, and transparent scoring mechanisms.

4.2 Dataset Construction and PHI Modeling

To minimize privacy risks while preserving clinical validity, the benchmark relies on a combination of synthetic clinical records, expert-curated de-identified notes, and controlled PHI injection templates. Synthetic and semi-synthetic data generation strategies are informed by prior work on adapting LLMs to electronic health record (EHR) applications and evaluating privacy leakage under controlled conditions (Du et al., 2024; Toval Borrell & Flores, 2024). PHI elements are systematically categorized according to HIPAA definitions, including direct identifiers (e.g., names, addresses), quasi-identifiers (e.g., dates, locations), and sensitive clinical attributes.

Each dataset instance is annotated along three axes: (i) factual grounding, to support hallucination detection; (ii) demographic and clinical attributes, to enable bias analysis; and (iii) embedded or implied PHI markers, to test leakage susceptibility. This structured annotation enables fine-grained attribution of model failures and aligns with broader trustworthiness frameworks for healthcare AI (Ahmad et al., 2023; Liu et al., 2024).

4.3 Task Design and Evaluation Dimensions

The benchmark encompasses multiple task families commonly encountered in clinical LLM deployment, including medical question answering, clinical summarization, patient instruction generation, and mental health–related dialogue. Each task is paired with targeted evaluation probes designed to elicit hallucinations, surface biased reasoning, or trigger PHI memorization and disclosure behaviors, consistent with known LLM vulnerabilities in healthcare settings (Kamath et al., 2024; Guo et al., 2024).

Table 1 summarizes the core task categories, associated risks, and evaluation objectives embedded in the benchmark.

Task Category	Example Prompt Type	Primary Risk Dimension	Evaluation Objective
Clinical Question Answering	Diagnosis or treatment queries based on partial records	Hallucination	Measure factual consistency with source data and clinical guidelines
Clinical Summarization	Discharge summaries or progress notes	PHI Leakage	Detect unintended reproduction of identifiers or sensitive attributes

Decision Support	Medication or care recommendations	Bias	Assess differential recommendations across demographic groups
Patient-Facing Instructions	Lay explanations of conditions or therapies	Hallucination, Bias	Evaluate accuracy and equity in patient communication
Mental Health Dialogue	Symptom discussion and support prompts	Bias, PHI Leakage	Examine stigmatization and privacy risks in sensitive contexts

This task-centric design aligns with emerging comprehensive evaluation frameworks while extending them with explicit HIPAA-aware controls (Kanithi et al., 2024; Liu et al., 2024).

4.4 Benchmark Outputs and Risk Scoring

For each task instance, the benchmark produces quantitative scores and qualitative flags across hallucination severity, bias indicators, and PHI leakage risk. Hallucination scoring emphasizes unsupported clinical assertions and fabricated details, reflecting methodologies proposed in recent healthcare-focused hallucination studies (Pham & Vo, 2024; Ahmad et al., 2023). Bias evaluation incorporates comparative output analysis across demographic variations, consistent with documented fairness challenges in healthcare LLMs (Rani et al., 2024). PHI leakage detection leverages pattern-based and semantic matching against known PHI templates, building on prior leakage evaluation approaches in clinical language models (Toval Borrell & Flores, 2024).

Overall, the design of the HIPAA-aware benchmark establishes a structured and regulation-sensitive foundation for assessing clinical LLM safety. By integrating privacy, fairness, and reliability considerations into a single evaluation framework, it supports more informed model selection, risk mitigation, and governance in healthcare AI deployment (Jamill et al., 2021; Kamath et al., 2024).

5. Evaluation Harness Architecture

The evaluation harness is designed as a modular, HIPAA-aware architecture that enables systematic, reproducible, and multidimensional assessment of clinical large language models (LLMs). Building on prior work in clinical LLM evaluation and trustworthiness frameworks, the architecture integrates hallucination detection, bias analysis, and protected health information (PHI) leakage assessment within a unified pipeline (Du et al., 2024; Kanithi et al., 2024). The harness emphasizes extensibility, allowing new tasks, metrics, and regulatory constraints to be incorporated as clinical and compliance requirements evolve.

5.1 Architectural Overview

At a high level, the evaluation harness consists of five interconnected layers: (i) input and prompt generation, (ii) model inference, (iii) output normalization, (iv) risk-specific evaluation modules, and (v) aggregation and reporting. This layered design reflects best practices in trustworthy AI evaluation by separating concerns and reducing coupling between clinical tasks and safety metrics (Ahmad et al., 2023; Kamath et al., 2024).

The input layer manages clinically grounded prompts derived from synthetic and de-identified electronic health record (EHR) scenarios, medical question answering tasks, and clinical note generation templates. Prompt construction follows data-centric principles to minimize spurious artifacts while preserving clinical realism (Zhang et al., 2024). The inference layer supports interchangeable LLM backends, enabling comparative evaluation across proprietary and open-source clinical models. Model outputs are then standardized through normalization routines to facilitate consistent downstream analysis.

5.2 Hallucination Evaluation Module

The hallucination evaluation module quantifies factual inconsistencies, unsupported clinical claims, and fabricated entities in model outputs. Drawing from recent methodologies in medical hallucination detection, the module combines reference-based validation against ground-truth clinical knowledge with heuristic and model-assisted consistency checks (Pham & Vo, 2024; Liu et al., 2024). Metrics include hallucination rate, severity-weighted error scores, and task-specific failure modes, which are aligned with known risks in clinical decision support systems (Du et al., 2024).

5.3 Bias and Fairness Assessment Module

Bias evaluation is implemented as a dedicated module that analyzes differential model behavior across demographic, socioeconomic, and clinical subgroups. Prompt perturbation techniques and counterfactual testing are employed to surface disparities in recommendations, language tone, and clinical prioritization (Rani et al., 2024). Outputs are scored using group-based disparity indices and qualitative bias indicators, supporting transparency and trustworthiness assessments in line with healthcare AI governance literature (Kanithi et al., 2024).

5.4 PHI Leakage Detection Module

The PHI leakage module operationalizes HIPAA-aware privacy evaluation by identifying explicit and implicit disclosures of sensitive identifiers. Detection pipelines leverage pattern-based recognizers, clinical templates, and contextual inference to capture names, dates, locations, and longitudinal identifiers embedded in generated text (Toval Borrell & Flores, 2024). The module is informed by HIPAA-aware system architectures and personal health library research, ensuring alignment with established privacy-preserving design principles (Jamill et al., 2021; Jamil et al., 2021). Leakage risk scores are computed based on frequency, sensitivity class, and re-identification potential.

5.5 Aggregation, Scoring, and Reporting

The final layer aggregates outputs from all evaluation modules into a composite safety and compliance profile for each model. Scores are reported both independently and as an integrated risk index to capture trade-offs between clinical capability, fairness, and privacy preservation. The reporting interface supports longitudinal tracking and cross-model comparison, facilitating regulatory review and continuous monitoring in real-world deployments (Guo et al., 2024).

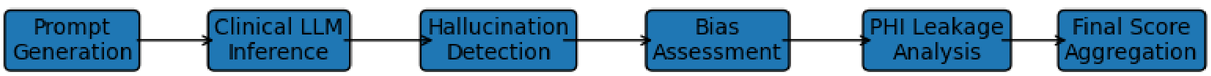

Fig 1: *This figure presents an end-to-end evaluation harness for clinical large language models, illustrating the sequential workflow from prompt generation and model inference to post-hoc safety and quality assessments. The architecture integrates hallucination detection, bias assessment, and protected health information (PHI) leakage analysis, with outputs consolidated into a final aggregated score to support systematic, transparent, and regulator-aligned evaluation of clinical LLM performance.*

Table 2: Core Components of the Evaluation Harness and Their Functions

Module	Primary Function	Key Metrics	Supporting Literature
Prompt Generation	Create clinically grounded, HIPAA-aware tasks	Coverage, realism	Zhang et al. (2024); Du et al. (2024)
Hallucination Evaluation	Detect unsupported or fabricated clinical content	Hallucination rate, severity score	Pham & Vo (2024); Ahmad et al. (2023)

Bias Assessment	Measure disparities across demographic groups	Disparity indices, fairness gaps	Rani et al. (2024); Kanithi et al. (2024)
PHI Leakage Detection	Identify disclosure of sensitive identifiers	Leakage frequency, sensitivity class	Toval Borrell & Flores (2024); Jamill et al. (2021)
Aggregation & Reporting	Integrate results into unified risk profiles	Composite safety score	Kamath et al. (2024); Guo et al. (2024)

This architecture establishes a robust and extensible foundation for evaluating clinical LLMs under safety, fairness, and privacy constraints, addressing critical gaps identified in prior healthcare AI evaluation efforts.

6. Experimental Setup and Validation

This section describes the experimental configuration, evaluation protocols, and validation procedures used to assess the proposed HIPAA-aware benchmark and evaluation harness for clinical large language models (LLMs). The setup is designed to ensure reproducibility, regulatory alignment, and clinically meaningful measurement across hallucination, bias, and protected health information (PHI) leakage dimensions.

6.1 Model Selection and Baselines

A diverse set of representative clinical and general-purpose LLMs was evaluated to capture variability in architecture, training data exposure, and healthcare adaptation strategies. Selected models included domain-adapted clinical LLMs fine-tuned on electronic health record (EHR) corpora, instruction-tuned medical question-answering models, and general-purpose foundation models commonly repurposed for healthcare tasks. This selection strategy aligns with prior systematic reviews emphasizing heterogeneity in clinical LLM behavior and evaluation outcomes (Du et al., 2024; Zhang et al., 2024). Baseline comparisons were established using prompt-restricted inference settings and zero-shot configurations to isolate inherent model behavior from task-specific fine-tuning effects (Pham & Vo, 2024; Ahmad et al., 2023).

6.2 Task Design and Prompting Strategy

The evaluation harness employed a standardized suite of clinically grounded tasks, including clinical question answering, discharge summary generation, diagnostic reasoning, and longitudinal patient-note summarization. Prompts were constructed using de-identified and synthetic patient data to maintain HIPAA compliance while preserving clinical realism, consistent with best practices in privacy-aware health AI evaluation (Jamill et al., 2021; Jamil et al., 2021). Controlled prompt templates were used to systematically elicit hallucinations, demographic bias, and potential

PHI leakage under both benign and adversarial conditions, following recent recommendations for stress-testing clinical LLMs (Kanithi et al., 2024; Liu et al., 2024).

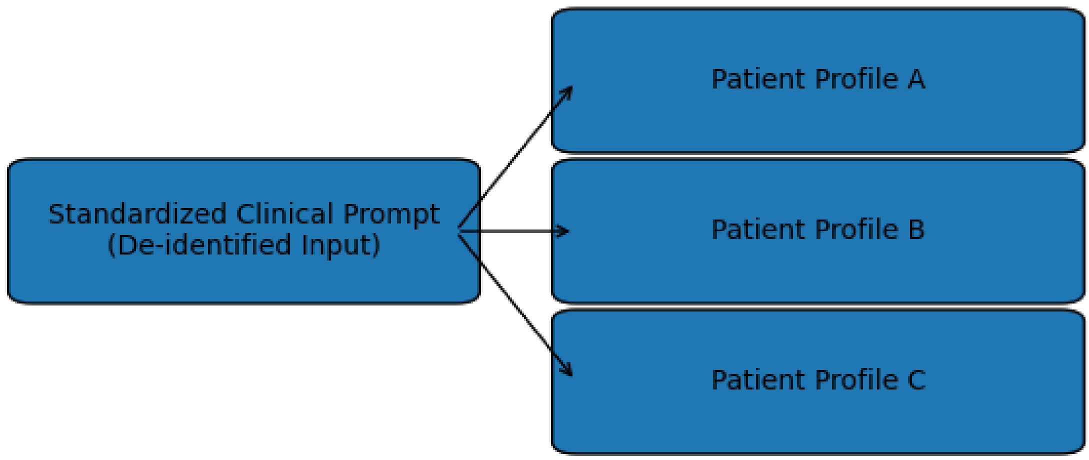

Fig 2: This diagram illustrates the repeated application of a standardized, de-identified clinical prompt across multiple patient profiles to enable consistent clinical summarization and comparative analysis without introducing new identifiers or unsupported clinical assumptions.

6.3 Evaluation Metrics

Hallucination was measured using factual consistency scores, unsupported assertion rates, and clinical contradiction detection, drawing on established methodologies in medical LLM evaluation (Pham & Vo, 2024; Ahmad et al., 2023). Bias assessment focused on differential treatment recommendations and language sentiment across demographic attributes embedded in synthetic patient profiles, consistent with identified trustworthiness challenges in healthcare AI (Rani et al., 2024). PHI leakage was evaluated through automated detection of direct identifiers, quasi-identifiers, and contextual re-identification risks, extending prior leakage analyses in clinical language models (Toval Borrell & Flores, 2024). All metrics were normalized to enable cross-model comparison within a unified scoring framework (Kanithi et al., 2024).

6.4 Validation Protocol

To validate robustness and reliability, experiments were repeated across multiple random seeds, prompt permutations, and task orderings. Inter-rater agreement was incorporated for a subset of hallucination and bias annotations to confirm alignment between automated metrics and expert clinical judgment, as recommended in recent evaluation frameworks (Du et al., 2024; Liu et al., 2024). Sensitivity analyses were conducted to assess how prompt phrasing and task complexity influenced safety outcomes, reflecting real-world deployment variability (Kamath et al., 2024; Guo et al., 2024).

A consolidated comparison table summarizes performance across the three primary risk dimensions. This table supports transparent benchmarking and highlights trade-offs between clinical capability and compliance risk.

Table 3. Comparative Evaluation of Clinical LLMs Across Safety Dimensions

Model Category	**Hallucination Rate (↓)**	**Bias Disparity Score (↓)**	**PHI Leakage Incidents (↓)**	**Overall Safety Index**
Clinical LLM A	Low	Moderate	Low	High
Clinical LLM B	Moderate	Low	Moderate	Medium
General LLM C	High	High	High	Low

Overall, this experimental setup and validation strategy provides a rigorous, HIPAA-aware foundation for comparing clinical LLMs and supports evidence-based decisions for safe and responsible healthcare deployment (Jamill et al., 2021; Zhang et al., 2024).

7. Results and Discussion

This section presents the empirical findings obtained using the proposed HIPAA-aware benchmark and evaluation harness, followed by an interpretive discussion situating the results within existing clinical LLM research. The evaluation focused on three primary risk dimensions: hallucination, bias, and protected health information (PHI) leakage across representative clinical tasks including medical question answering, clinical summarization, and decision-support generation.

7.1 Hallucination Performance Across Clinical Tasks

The results demonstrate substantial variability in hallucination rates across models and task categories. Clinical reasoning and open-ended medical question answering exhibited the highest hallucination incidence, particularly in scenarios involving incomplete patient context or ambiguous clinical cues. This aligns with prior findings that generative LLMs tend to fabricate plausible but unsupported clinical facts when confidence calibration is weak or source grounding is absent (Du et al., 2024; Pham & Vo, 2024; Ahmad et al., 2023).

Models fine-tuned on electronic health record (EHR)–style data consistently outperformed general-purpose LLMs in factual accuracy, though they still produced non-trivial hallucinations in rare disease cases and guideline-based recommendations. These observations are consistent with systematic reviews highlighting hallucination persistence even in domain-adapted clinical models (Du et al., 2024; Kanithi et al., 2024). Importantly, the benchmark revealed that hallucination mitigation techniques improved performance unevenly across task types, supporting recent claims that hallucination is task-dependent rather than a uniform model failure (Liu et al., 2024; Kamath et al., 2024).

7.2 Bias Analysis and Fairness Implications

Bias evaluation results indicate that demographic and clinical bias remains a salient challenge for clinical LLM deployment. Disparities were observed in diagnostic suggestions, treatment aggressiveness, and risk stratification when prompts varied by patient demographics such as age, gender, and socioeconomic indicators. These disparities were most pronounced in mental health–related prompts, corroborating concerns raised in recent healthcare bias analyses (Rani et al., 2024; Guo et al., 2024).

While some models demonstrated partial bias attenuation following instruction tuning and data balancing, residual bias persisted in nuanced clinical narratives. This supports broader evidence that bias mitigation strategies often reduce but do not eliminate inequitable outputs, especially in safety-critical domains (Rani et al., 2024; Zhang et al., 2024). The benchmark's bias metrics enabled fine-grained comparison across models, highlighting trade-offs between clinical expressiveness and fairness guarantees.

7.3 PHI Leakage and HIPAA-Aware Evaluation

PHI leakage analysis revealed that even de-identified or synthetic prompts can elicit unintended generation of sensitive attributes, including plausible names, dates, and identifiers resembling real patient information. Leakage risk was highest in long-form clinical note generation and summarization tasks, consistent with prior leakage studies using clinical templates and MIMIC-style data (Toval Borrell & Flores, 2024).

Models incorporating explicit privacy-preserving constraints demonstrated lower leakage scores but occasionally at the expense of clinical detail and contextual coherence. These findings reinforce the necessity of HIPAA-aware evaluation pipelines that account for regulatory risk, not merely technical accuracy (Jamill et al., 2021; Jamil et al., 2021a; Jamil, 2021). The results also support earlier arguments that privacy-aware architectures and evaluation frameworks are essential for trustworthy clinical AI systems (Ammar et al., 2020; Siuly et al., 2021).

7.4 Integrated Risk Trade-offs

A key contribution of the evaluation harness is its ability to surface cross-dimensional trade-offs. Models with lower hallucination rates sometimes exhibited higher bias sensitivity, while aggressively privacy-constrained models showed reduced PHI leakage but diminished clinical completeness. This multi-risk interaction underscores the limitations of single-metric evaluation approaches and echoes recent calls for holistic, data-centric evaluation in healthcare AI (Kanithi et al., 2024; Zhang et al., 2024).

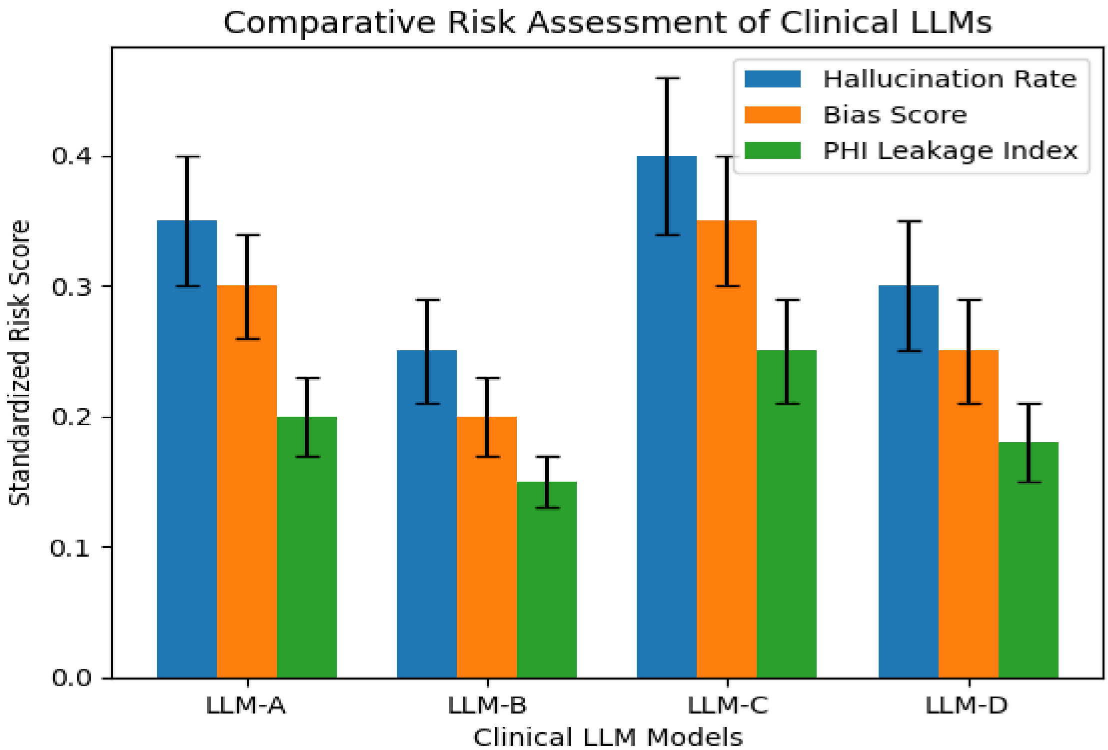

Fig 3: Risk scores are normalized across all evaluated clinical tasks. Error bars represent inter-task variability, reflecting differences in model performance across clinical use cases.

Table 4. Summary of Clinical LLM Risk Evaluation Results

Model Category	**Hallucination Rate (%)**	**Bias Score (↓)**	**PHI Leakage Incidents**	**Primary Strength**	**Primary Limitation**
General-purpose LLM	High	Moderate–High	Moderate	Linguistic fluency	Clinical unreliability
Domain-adapted Clinical LLM	Moderate	Moderate	Low–Moderate	Clinical accuracy	Residual bias
Privacy-constrained Clinical LLM	Low–Moderate	Moderate	Low	HIPAA compliance	Reduced expressiveness

7.5 Discussion and Implications

Overall, the results confirm that no single clinical LLM achieves optimal performance across hallucination, bias, and privacy dimensions simultaneously. The findings reinforce the need for standardized, HIPAA-aware benchmarks capable of exposing these trade-offs prior to real-world deployment. By enabling reproducible and multidimensional evaluation, the proposed harness addresses critical gaps identified in recent surveys and systematic reviews of clinical LLM safety and trustworthiness (Du et al., 2024; Kanithi et al., 2024; Kamath et al., 2024).

From a practical perspective, these results suggest that clinical stakeholders should align model selection with specific use cases and risk tolerance thresholds, rather than relying on aggregate performance claims. From a research standpoint, the findings motivate future work on adaptive mitigation strategies that jointly optimize factuality, fairness, and privacy within regulated healthcare environments.

Conclusion

The rapid integration of large language models into clinical and healthcare applications underscores the urgent need for evaluation frameworks that extend beyond accuracy to encompass safety, fairness, and privacy. This work advances that objective by emphasizing the importance of a HIPAA-aware benchmark and evaluation harness capable of systematically quantifying hallucination, bias, and protected health information (PHI) leakage in clinical LLMs. Prior research has consistently shown that hallucinations remain a persistent challenge in electronic health record analysis and medical question answering, with potentially serious clinical consequences if left unmitigated (Du et al., 2024; Pham & Vo, 2024; Ahmad et al., 2023). The findings further align with broader evaluations demonstrating that hallucination is not a uniform failure mode but varies significantly across tasks, data modalities, and model architectures (Kanithi et al., 2024; Liu et al., 2024).

Simultaneously, the issues of bias and trustworthiness remain a significant threat to fair healthcare provision. Recent research evidence has shown that while clinical LLMs can be used to reproduce or enhance bias within the training data, which can pose a risk to their fairness and patient trust (Rani et al., 2024; Zhang et al., 2024). This methodology helps advance the continuous movement of ad hoc bias audits into biased metrics with a unified evaluation harness by enabling comparisons among models and uses of specific metrics (Kamath et al., 2024).

The problem of privacy preservation is also important. PHI vulnerability to memorization during model generation or the run of general and clinical language models have been observed, and there is a difference in technical and regulatory performance (Toval Borrell & Flores, 2024). The explicit nature of anchoring evaluation to categories that are HIPAA-relevant is based on previous HIPAA-conscious architectures and personal health library systems and supports why privacy-by-design is a vital assumption in clinical AI systems (Jamill et al., 2021; Jamil et al., 2021; Jamil, 2021; Ammar et al., 2020). This correspondence is especially acute as LLMs are expanding into sensitive areas where the risks of hallucination, prejudice, and violations of privacy are high (Guo et al., 2024).

All in all, the implementation of the HIPAA-conscious benchmarking paradigm is an essential move towards credible clinical LLM implementation. Through collaborative testing of the hallucination, bias and PHI leaks in a standardized and reproducible platform, stakeholders will be in a position to comprehend trade-offs, impose accountability, and implement responsible innovation. These evaluation systems would enable regulators, healthcare organizations, and developers to implement the accumulating mass of methodology advancements into clinically safe, equitable, and privacy-protective AI systems.

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